

Liquid fuels demand and supply model

Manish hand-back

6 October | Sourced data, explicit gaps and levers

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Five packages are back: three complete, two partial with stated gaps

Handback against the 6 October checklist | petrol and diesel; jet tracked separately

Work package	What came back	Status
1 Integrity	Every P1 flag traced to its source cell. Three extraction errors fixed, one of them ours (2013 sales read from a superseded sheet). Industry and agriculture baselines now sourced.	Complete; decisions with Nigel
2 Traceability	45 datasets mapped from publisher to original, extract and consuming function. The engine still reads six files, all from the Reatile workbook.	Complete
3 Provincial demand	Nothing official exists after 2023-Q1; the department's own schedule promised data to end-2024. Estimate for 2023-24: petrol within 2-3%, diesel 5-11%.	Partial: no later data exists
4 Fuel balance	SARS customs data fetched by script, 2010 to August 2026. Sales less net imports computed by product. A diesel gap of 3-4 bn litres a year is a hypothesis to test.	Partial: no product split or stocks
5 Driver evidence	All seven categories sourced or marked. Prices to October 2026, quarterly GDP, mining, manufacturing, freight and CPI series added.	Complete; open rows listed

155 tests pass; lfm check passes with 52 open exceptions; both scenarios run. No change is on main.

Five input changes are applied on the branch; one moves model results

Approved by Manish, awaiting Nigel | register reconciled, every changed row marked unreviewed

Change	Evidence	Effect on the model
2013 national sales: diesel 11.89 to 12.14, petrol 11.15 to 11.44 bn litres	Department's published sheet, both district workbooks and the Reatile history agree	None: reference data
Refinery file: 948 rows to 348	Production_High range ran to row 13,102 and swept in seven other tables	None: supply identical every year
Jet history: 72 rows to 57	Extraction ran into three regional tables below the national one	None
Industry baseline 2.5 to 1.50 bn litres; agriculture 0.7 to 1.06	Department energy balance, 2021. Placeholders had no source	Diesel demand falls 0.63 bn litres a year
New reference data: SARS trade, operator output, prices, quarterly GDP, Stats SA monthly, power diesel	Each with an evidence record and register rows	None: not read by the engine

Left unchanged for Nigel: marine baseline, elasticities, road parameters, 2014 and 2018 provincial treatment.

Customs data settles 2024 diesel imports and shows where fuel enters

SARS trade statistics, chapter 27 fuel lines | litres from 2014; fetched by script

2024, bn litres	SARS	FIASA	Trade report
Diesel imports	10.793	14.793	10.8
Petrol imports	4.001	4.000	4.0
Diesel exports	0.796	0.821	0.79
Petrol exports	0.920	0.939	0.92

Imports cleared at	2024	Share
Durban	11.84	80%
Cape Town	1.09	7%
Mossel Bay	0.62	4%
East London	0.53	4%
Richards Bay	0.39	3%

01 | FIASA's figure is a misprint

FIASA's 14 793 is exactly 4 000 million litres above customs. In 2014-17, 2020 and 2025 the two agree to within 12 million litres.

02 | Durban clears four-fifths of imports

Petrol plus diesel, by customs office. This is a public proxy for entry port, not a terminal record. 98% of diesel imports arrive by sea.

03 | Botswana is sourcing less through South Africa

Its recorded imports from South Africa fell from 1.02 to 0.63 bn litres between 2023 and 2024; SARS exports show the same direction.

Four indicative comparisons point to a possible diesel gap; none is matched accounting

Hypothesis to test: 3-4 bn litres a year | not added to demand | production and stocks unresolved

Route	What it shows	Size
Product balance	Sales less net imports is 0.9-1.7 bn litres for diesel in 2022-24, beside 4.6-6.0 bn for petrol. It is a balancing requirement, not output. The plants made 0.85 litres of diesel per litre of petrol in 2017-19; that ratio need not hold now.	2.2-3.7 bn on that ratio
Operators' output	Secunda, Natref and Astron report 13-14 bn litres of all refined products; the balance needs 8.7 (2023) and 10.6 (2024). Fiscal against calendar years; conversions assumed.	4.4 and 3.3 bn
Customs, both sides	SARS confirms the import and export figures. Nine neighbours report importing 1.8-2.0 bn litres from South Africa against 1.6-1.9 recorded as exported.	Exports do not explain it
Fuel levy volumes	SARS: levy declared on 24 bn litres in the eleven months to February 2024, against about 20 bn of recorded sales for roughly the same months. The period is assumed.	About 3.9 bn

One reading is that the department's sales series undercounts. Stock changes, plant product mix and mismatched periods could give the same differences. It stays a hypothesis until production and stocks are matched.

Fuel prices have risen sharply in 2026 and the model has no price response

Regulated inland prices, rand per litre | department to February 2026, Central Energy Fund sheets after

Effective	4 Feb	4 Mar	1 Apr	6 May	3 Jun	1 Jul	5 Aug	2 Sep	7 Oct
Petrol 95, retail	20.10	20.30	23.36	26.63	28.06	26.10	25.58	26.92	30.25
Diesel 0.05%, wholesale	17.92	18.54	25.91	31.18	27.93	24.79	26.17	29.11	31.95

01 | Diesel is up 78% and petrol 50% since February

Our price series stopped in April 2024 this morning. It now runs monthly from January 2011 to October 2026.

02 | One South African study gives the response

Boshoff (2012): a 10% rise in the real price lowers petrol demand by about 5% and diesel by about 1% in the long run.

03 | Coastal prices are no longer published

Coastal diesel has no value from December 2025; coastal petrol and paraffin none from March 2026. The fund prints Gauteng prices only.

04 | Power diesel: 0.31 litres per kWh

Three years of reported burn against turbine output agree. The workbook's efficiency implies 0.244, a fifth too low. Adoption is left for Nigel.

The seven levers each have evidence and a proposed base case; none is calibrated

Draft for discussion | base case = things continue as they perform today

Lever	Evidence held	Proposed base case	Needed to reach litres
Road to rail	Rail 214 Mt (2019), 156 (2022), 168 (2025); road 895 to 975. Target 250 Mt by 2029/30	Rail holds its present share	Haul distance; diesel per tonne-km
Electric vehicles	2.8% of 2025 sales against a quoted 20% target; 0.2% battery electric. Peers: Brazil 6%, Southeast Asia near 20%	Continues from actual sales	AIA's view of the path; fleet turnover
Grid power	Turbine output FY2022-26; 0.31 litres per kWh	Eskom's own plan	Decision on the factor
Industry	Mining and manufacturing volumes 7% below 2019, flat	Flat, then with GDP	Litres per unit of output
Agriculture	Value added to 2025; diesel use to 2021	With sector value added	Choice of driver
Vehicle fleet	12.6 million vehicles, growing 1.4% a year	Present growth rate	Fuel split by class; distance after 2014
Growth	Treasury 1.6%, 1.8%, 2.0%; then 2%	As set on 5 October	Wiring into the model

Five further levers are proposed, mostly where the model is silent

Additions to the seven | two do not sit on a high-low demand axis and would be shown as variants

Proposed lever	Why it belongs	Proposed base case
Fuel price level	Diesel up 78% and petrol 50% since February; the model has no price response	Prices settle at a stated level
Private backup generation	The power lever covers grid turbines only. SARS attributes a 3 bn litre fall in levy volumes to increased energy availability	No return of load-shedding
Neighbours' sourcing through South Africa	Throughput for Vopak, not South African demand. Botswana's imports via South Africa fell 38% in a year	Present volumes continue
Refinery events	Petrol and diesel must meet 10 ppm sulphur from 1 July 2027. Natref does not yet comply; Astron is investing R6 bn. A miss raises imports	Three plants run and comply
Passengers returning to rail	503 million journeys in 2010, 19 million in 2022, 103 million in 2025. Probably small	Recovery continues

Grid turbines and backup generation move together; so do electric vehicles and fuel prices. A matrix should not treat them as independent.

Decisions for Nigel before integration and lever calibration

Each has evidence in the notes under workstreams/WS1_data_validation and WS2_model_development

Decision	Evidence in brief	Suggestion
Confirm the five applied changes	Slide 3; flag log	Accept
2024 diesel imports and how to report the diesel gap	Customs 10.793 bn litres; four routes to 3-4 bn	Use 10.79; report the gap as a hypothesis
Power diesel factor	0.31 litres per kWh in three years against 0.244	Adopt 0.31
Marine baseline	2.2 bn litres equals the department's 2007 figure; trade press now about 1 bn	Hold until an official figure
Elasticities and road parameters	One study for diesel as a whole; 2018 paper for vehicles	Keep 1.0 for industry; review the vehicle mapping
Provincial 2023-24 in the pack; 2014 and 2018 treatment	Estimate and back-test; original cells	Petrol only, labelled; national stands
Sales less net imports after 2021; 2019-20 balances for trade	Balance note	Not a measure of output; stop using them
Remaining originals; register layout	16 datasets held locally	Nigel's call

Review the branch and decide

manish-branch | checks pass | nothing
merged to main

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